Huawei Access Node BTM-ACC-23 QoS Congestion Event ALARM_7008 Warning Severity SOP

Node: BTM-ACC-23 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7008

Description

This Standard Operating Procedure (SOP) addresses the occurrence of QoS congestion events on the Huawei access node BTM-ACC-23, specifically ALARM_7008 warning severity. The ALARM_7008 alarm code is indicative of microbursts, interface tail drops, or QoS queue starvation, which can cause packet loss, jitter, and increased latency. This failure domain is particularly pertinent on the BTM-ACC-23 node due to its role as an aggregation point for various services, resulting in an aggregated traffic load that may overwhelm the node's capabilities. In the context of Huawei equipment, the egress queues of the access node are managed by the Quality of Service (QoS) scheduler, which allocates bandwidth and sets queue thresholds to prevent congestion. However, under heavy load conditions, the QoS scheduler may not always be able to keep up with the demand, leading to congestion events and subsequent alarm generation. This SOP aims to provide a structured approach to diagnosing and resolving QoS congestion events on the BTM-ACC-23 node, ensuring minimal service disruption and optimal network performance.

Pre-requisite Checks

- Verify that the BTM-ACC-23 node is accessible via the Huawei CLI.

- Check the power and environmental status of the node and surrounding equipment.

- Verify the status of redundant links and ensure they are functioning correctly.

- Dispatch remote hands to BTM Layout Aggregation, if necessary, to perform physical checks and ensure access to the node.

Potential Root Causes

Cause	Probability
Insufficient egress queue bandwidth allocation	High
QoS scheduler misconfiguration	Medium
Interface hardware limitations	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue-profile profile-name, Expected Output: Queue profile name with bandwidth allocation settings, Description: This command displays the bandwidth allocation settings of the specified queue profile, which is used to configure the egress queues of the access node.
```

```
Command: display qos profile profile-name, Expected Output: QoS profile name with scheduler settings, Description: This command displays the scheduler settings of the specified QoS profile, including the allocation of bandwidth to egress queues.

Command: display interface interface-name, Expected Output: Interface name with hardware specifications, Description: This command displays the hardware specifications of the specified interface, including its maximum transmit rate and buffer size.
```

Resolution Steps

Step 1: Step: Increase egress queue bandwidth allocation, Command: queue-profile profile-name bandwidth-allocation, Description: Increase the bandwidth allocation of the specified queue profile to prevent congestion events on the egress queues.

Step 2: Step: Reconfigure QoS scheduler, Command: qos-profile profile-name scheduler-settings, Description: Reconfigure the QoS scheduler to allocate bandwidth to egress queues more efficiently, preventing congestion events.

Step 3: Step: Replace interface hardware, Command: replace interface interface-name, Description: Replace the interface hardware with a newer model that has improved performance characteristics, such as increased transmit rate and buffer size.

Post-Check Verification

```
Command: display queue-profile profile-name, Expected Output: Queue profile name with updated bandwidth allocation settings, Description: Verify that the bandwidth allocation settings of the specified queue profile have been updated correctly.

Command: display qos profile profile-name, Expected Output: QoS profile name with updated scheduler settings, Description: Verify that the scheduler settings of the specified QoS profile have been updated correctly.

Command: display interface interface-name, Expected Output: Interface name with updated hardware specifications, Description: Verify that the hardware specifications of the specified interface have been updated correctly.
```

Roll-back Procedures

- Step: Restore original queue profile bandwidth allocation, Command: queue-profile profile-name restore, Description: Restore the original bandwidth allocation settings of the specified queue profile if the resolution steps fail to resolve the issue.

- Step: Restore original QoS profile scheduler settings, Command: qos-profile profile-name restore, Description: Restore the original scheduler settings of the specified QoS profile if the resolution steps fail to resolve the issue.

- Step: Restore original interface hardware, Command: replace interface interface-name restore, Description: Restore the original interface hardware if the resolution steps fail to resolve the issue.